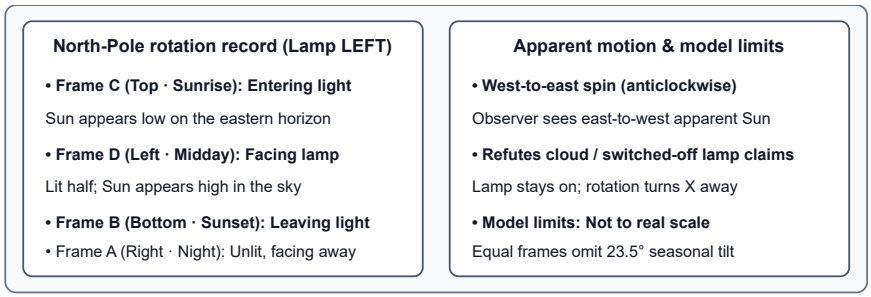
Reference sheets

Reference A · Four-frame North-Pole rotation record (Fixed lamp on LEFT)

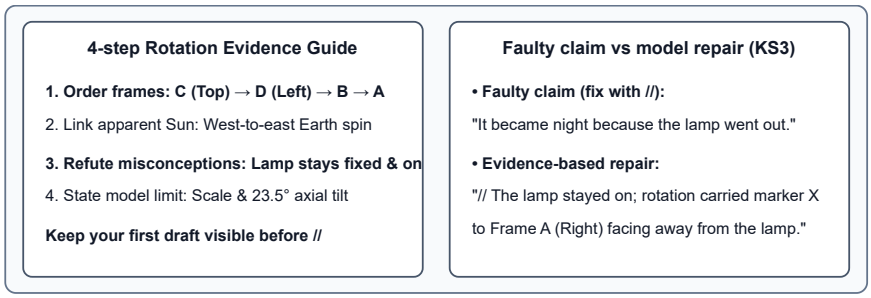


North-Pole rotation record. It links Frames C, D, B, and A to apparent Sun motion and limits.

Reference only. Do not write on this sheet. Use these supplied observations to support your rotation explanation.

Ordered frame & position.	Illumination state of marker X.	Apparent Sun position from marker X.	Rotation clue & model note.
1. Frame C · Top (Sunrise).	Entering the lit left half from shadow.	Sun appears low on the eastern horizon.	Start of cycle (0°); Earth spins anticlockwise (west to east).
2. Frame D · Left (Middle of day).	Facing the fixed lamp directly in the lit half.	Sun appears high across the sky.	After 1st quarter turn (90°); lamp stays fixed and on.
3. Frame B · Bottom (Sunset).	Leaving the lit left half into shadow.	Sun appears low on the western horizon.	After 2nd quarter turn (180°); observer turns away from lamp.
4. Frame A · Right (Night).	Facing away from the lamp in the unlit right half.	Sun is below the local horizon (darkness).	After 3rd quarter turn (270°); 4th turn returns X to Top (360°).

Reference B · Refuting misconceptions, apparent motion, and model limits



Four-step Rotation Evidence guide. It shows frame ordering, apparent motion, refuting cloud claims, and limits.

Evidence step.	Sentence stem for your book.	Key Stage 3 check.
1. Order the 24-hour cycle.	From sunrise, marker X moves through C (Top), D (Left), B (Bottom), and A (Right).	Did you track the same marker back to its starting point?
2. Explain apparent Sun motion.	Earth rotates west to east, so the Sun appears to move east to west.	Did you distinguish observer viewpoint from real solar motion?
3. Refute 'clouds/lamp off' claims.	No clouds are present and the lamp stays on. Rotation turns X into shadow.	Did you use Frame D and Frame A to prove orientation changed?
4. State one honest model limit.	// Limit: size and distance are not to scale. Equal frames omit 23.5° tilt.	Did you state what equal quarter turns cannot show?

Made by Matt

Tier pack · Supported

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Look at Box 1 on the left of the arrival drawing. What does rotation mean: turning on an axis, or switching off a lamp?

Help. Look at the spinning globe in Box 1.

2. Look at Box 1 again. Marker X faces the fixed lamp on the left. Is marker X in day or night?

Help. Check the lit left half in Box 1.

3. Look at Box 2 in the middle of the drawing. The lamp stayed on, and marker X turned into shadow. What moved: the Earth model or the lamp?

Help. Check Box 2 in the middle.

4. Look at Box 3 on the right of the drawing. What completes one full rotation cycle: returning to the start, or stopping halfway?

Help. Check the four-step loop in Box 3.

You do tasks

1. Use Reference A. Order Frames C, D, B, and A starting at sunrise (Top). Match each frame to entering light, facing light, leaving light, or facing away. **Staff I ☐ S ☐ C ☐**

Help. Check the four-row table on Reference A.

2. Use Reference A. Explain why Frame D is day and Frame A is night. Complete: 'Marker X has night because Earth ___ and X faces ___ from the lamp.' **Staff I ☐ S ☐ C ☐**

Help. Use the words rotates and away.

3. Use Reference B. Refute the claim 'Clouds make night.' Then add one // improvement after adult feedback. **Staff I ☐ S ☐ C ☐**

Help. Check the lamp and cloud evidence on Reference B.

Optional participation-loop prompt. Share your four-frame evidence and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Look at the North-Pole model. Which frame is in night: Frame D (Left) or Frame A (Right)?

2. Does the model need clouds to make night? Give one clue from the model.

3. How do we mark an improvement in our book after feedback?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Supported

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt